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Collatz Conjecture Tools - PHP Execution Manual

1. Introduction

This manual explains how to run the following PHP programs related to the Collatz Conjecture.

Program Name	Description
Collatz_Launcher.php	(Recommended) Unified launcher combining all 5 tools below into a single application with a menu
Counting_by_Length.php	Counts converged/unconverged Parity Vectors (PV) by length
Counting_by_Hamming_Weight.php	Counts converged/unconverged PVs by Hamming weight
Collatz_Calculator.php	Displays the Collatz operation process for any natural number
Collatz_PV_Graph_from_String.php	Graphical display of an input Parity Vector
Collatz_PV_Graph_from_Number.php	Graphical display of the PV generated from an input natural number N

All of these are server-side PHP programs accessed through a web browser.

Tip: Collatz_Launcher.php combines all five tools into one program with a menu bar at the top, so you only need to access a single URL and use one browser tab to reach every tool. The five individual programs remain available separately if you prefer to run them independently.

2. Prerequisites

2.1 Required Software

To run PHP, you need a web server and a PHP runtime environment. The easiest way is to use **XAMPP**.

Software	Role	Source
XAMPP	All-in-one package including Apache (web server), PHP, and MySQL	https://www.apachefriends.org/

Confirm that XAMPP includes PHP and the GMP extension (used for accurate large-integer arithmetic). If it is not included, see “2.3 Checking the GMP Extension” below.

2.2 Installing XAMPP

1. Download XAMPP from the site above (available for Windows / Mac / Linux).
2. Run the installer and follow the on-screen instructions.
3. After installation, launch the “XAMPP Control Panel”.
4. Click the Start button next to **Apache**.

If it turns green and shows Running, the server has started successfully.

2.3 Checking the GMP Extension

These programs rely on PHP’s **GMP extension** for accurate handling of large integers.

How to check:

1. Open the `php.ini` file inside your XAMPP installation folder.
 - Windows example: `C:\xampp\php\php.ini`
2. Find the following line:
`;extension=gmp`
3. Remove the leading `;` (semicolon) and save the file.
`extension=gmp`
4. In the XAMPP Control Panel, stop Apache once and then start it again.

Quick verification test:

Create a file named `phpinfo.php` with the following content and open it in your browser.

```
<?php phpinfo(); ?>
```

If a “gmp” section appears on the page, the extension is enabled.

3. Setting Up the Programs

3.1 Installation Location

With XAMPP, the web server only recognizes files placed in a specific folder.

Windows example:

`C:\xampp\htdocs\`

Mac example:

`/Applications/XAMPP/htdocs/`

Create a subfolder of your choice (e.g., `collatz`) inside `htdocs`, and place the PHP files there.

The `data/` folder is created automatically when any program is run. There is no need to create it manually. If you only plan to use `Collatz_Launcher.php`, the five individual program files are not required — the launcher contains all of their functionality internally.

4. How to Run the Programs

4.1 Recommended: Using the Unified Launcher

The simplest way to use all five tools is to access the launcher, which provides a menu bar for switching between them — no need to remember five different URLs.

1. Confirm that **Apache** is running in the XAMPP Control Panel
2. Open your web browser
3. Enter the following address:

`http://localhost/collatz/Collatz_Launcher.php`

You will see a top menu bar with the following items. Click any item to switch tools instantly:

- **Home**
- **1. Counting by Length**
- **2. Counting by Hamming Weight**
- **3. Collatz Calculator**
- **4. PV Graph (from PV string)**
- **5. PV Graph (from Number)**

All result files from any tool are saved together in the same `data/` folder next to `Collatz_Launcher.php`.

4.2 Alternative: Accessing Individual Programs

If you prefer to use a single tool on its own, you can access each program directly using the following address format:

`http://localhost/collatz/program_name.php`

4.3 Examples

Counting by Length:

`http://localhost/collatz/Counting_by_Length.php`

Counting by Hamming Weight:

http://localhost/collatz/Counting_by_Hamming_Weight.php

Collatz Calculator:

http://localhost/collatz/Collatz_Calculator.php

PV Graph from a String:

http://localhost/collatz/Collatz_PV_Graph_from_String.php

PV Graph from a Number:

http://localhost/collatz/Collatz_PV_Graph_from_Number.php

Once the page is displayed, simply enter values into the form and press “Execute”.

Note: You do not need to access the individual programs if you are using `Collatz_Launcher.php` — it already includes all of their functionality. The individual programs are only needed if you specifically want to use one tool by itself.

5. Stopping the Program

PHP programs themselves have no “stop” operation. Simply close the browser tab to end your session.

If you want to stop the web server (Apache) itself, click the Stop button next to **Apache** in the XAMPP Control Panel.

6. Overview of Each Program

6.0 Collatz_Launcher.php (Unified Launcher)

- **Input/Output:** Combines all five tools below (6.1–6.5) into a single application
- **Navigation:** Use the top menu bar to switch between tools; each menu item loads via `?tab=...` in the URL
- **Storage:** All result files from any tool are saved in the same shared `data/` folder
- **Recommended** as the default way to use this tool suite

6.1 Counting_by_Length.php

- **Input:** Start length (A), end length (B), ratio precision
- **Output:** A table summarizing converged/unconverged PV counts for each length k
- Result files are saved as `.txt` in the `data/` folder

6.2 Counting_by_Hamming_weight.php

- **Input:** Start Hamming weight (A), end Hamming weight (B)
- **Output:** A table summarizing converged/unconverged PV counts for each Hamming weight d
- Result files are saved as .txt in the data/ folder

6.3 Collatz_Calculator.php

- **Input:** Any natural number (2 or greater)
- **Output:** Each step of the Collatz operation, the Glide value, and the Parity Vector
- If file output is selected, results are saved as .csv in the data/ folder

6.4 Collatz_PV_Graph_from_String.php

- **Input:** A Parity Vector (a string of 0s and 1s)
- **Output:** A graphical matrix of the PV path and a detailed information table
- Detailed result files are saved as .html in the data/ folder

6.5 Collatz_PV_Graph_from_Number.php

- **Input:** Any natural number N (2 or greater)
- **Output:** Automatically computes the PV from N up to its Glide point and displays it graphically
- Uses the same graphical style as Collatz_PV_Graph.php

7. Troubleshooting

Symptom	Cause and Solution
Page does not load (connection error)	Apache is not running. Click Start in the XAMPP Control Panel
Error: Call to undefined function gmp_init()	The GMP extension is disabled. See Section 2.3 to enable it
404 Not Found	The folder or file name in the URL is incorrect. Check the file placement inside htdocs
Garbled characters	Confirm your browser's character encoding is set to UTF-8 (usually automatic)
Calculation takes a very long time	The input range may be too large. Check the input field's upper limit (often 10000)
Warnings such as Warning: htmlspecialchars()	This is a character-encoding declaration issue; the programs covered in this manual have already been fixed
Apache fails to start because port 80 is in use	Another application (e.g., Skype) may be using port 80. Close it, or change the port number in XAMPP's settings

8. Frequently Asked Questions (FAQ)

Q. Do I need to start XAMPP every time I want to use a program? A. Yes. The PHP programs cannot be accessed unless Apache is running. You can also configure Apache to start automatically when your computer boots, via the XAMPP Control Panel settings.

Q. Where are the calculation result files saved? A. They are saved in the data/ folder located next to each program.

Q. Do I need an internet connection? A. No. Once XAMPP is installed, everything runs completely offline.

Q. Can I use multiple programs at the same time? A. Yes. Since PHP runs on a web server, there is no port conflict to worry about, unlike with Python. You can access multiple programs simultaneously simply by switching between browser tabs. With Collatz_Launcher.php, you can also just use the top menu bar to switch between tools within a single tab.

Q. Will this work the same way on other computers? A. Yes. As long as XAMPP is installed, it works the same way on Windows, Mac, or Linux.

9. Summary: Quick URL Reference

1. One-time only: install XAMPP and enable the GMP extension

2. Start Apache in the XAMPP Control Panel

3a. Recommended: access the unified launcher (all 5 tools, one menu)
`http://localhost/collatz/Collatz_Launcher.php`

3b. Alternative: access an individual program directly
`http://localhost/collatz/Counting_by_Length.php`
`http://localhost/collatz/Counting_by_Hamming_Weight.php`
`http://localhost/collatz/Collatz_Calculator.php`
`http://localhost/collatz/Collatz_PV_Graph_from_String.php`
`http://localhost/collatz/Collatz_PV_Graph_from_Number.php`

4. To stop (optional): Stop Apache in the XAMPP Control Panel

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